Inferring Semantic Social Networks from Scientific Texts: The Case of Astrobiology

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Introduction

Social network analysis (SNA) is a powerful methodology for understanding relationships between actors, whether they are individuals, organizations, or groups. SNA is widely used to study structures such as scholarly networks and scientific collaborations. By analyzing the nodes (actors) and edges (connections) of social networks, SNA provides insights into key actors, their relationships, and the structural dynamics of networks over time. In science studies, SNA has proven essential for exploring the social fabric of scientific communities. Studies have uncovered "hidden colleges" of researchers (Crane 1969; Zuccala 2006), traced the dynamics of collaboration (Tang et al. 2008), and illuminated the role of networks in discovery and career progression (Fortunato et al. 2018; Kong et al. 2019). Such analyses often depend on well-documented relational data, such as co-authorship, citations, or collaborations. This limits the scope of SNA, particularly in contexts where relational data are unavailable. Furthermore, SNA typically require supplementary metadata—for instance about scientific specialties—for interpretation.

Here we propose a novel approach to infer social networks directly from the textual data produced by actors, focusing on semantic similarities. This method is particularly relevant for contexts where actors' texts are available but explicit relational data are scarce or absent. By analyzing the semantic content of texts, we aim to uncover "hidden communities of interest" (HCoIs)—groups of actors connected by shared semantic content, even when their social ties are implicit or unknown. Our approach combines topic modeling and community detection to reconstruct actor networks based on the semantic similarity of their texts. This allows us to identify "semantic social networks" where the relationships between nodes emerge from textual similarities, rather than explicit relational data. Furthermore, interpretation of identified communities is facilitated by extracting topic profiles from the textual data. By extending this framework with diachronic analysis, we can also examine the temporal evolution and genealogies of these communities.

Semantic networks, which map relationships between terms based on their co-occurrence, have been extensively applied to study word associations and named-entity relations (Carley 1993; Diesner and Carley 2004). While these methods can be adapted to analyze actors' texts, our approach goes further by leveraging topic modeling to infer semantic similarities at the actor level. Similar efforts have been explored in directional social media networks (McCallum, Wang, and Corrada-Emmanuel 2007) and co-authorship data (Zhou et al. 2006), notably by adding prior constraining data on nodes, as in semi-supervised community detection algorithms and graph neural network approaches (Yang et al. 2014; Ye, Chen, and Zheng 2018). Our focus here is on inferring networks where relational data are absent.

Corpus and methods

Our study focuses on a corpus of 3,698 articles from three key astrobiology journals: *Origins of Life and Evolution of Biospheres*, *Astrobiology*, and *International Journal of Astrobiology*. These journals represent critical vectors for the institutionalization of astrobiology as a discipline (Dick and Strick 2004). By analyzing articles from 1968 to 2020, we aim to identify communities of researchers based on shared research interests, as reflected in the semantic content of their publications. The proposed approach comprises five main steps: (1) Topic Modeling—We applied Latent Dirichlet Allocation (LDA) to extract probabilistic topic distributions for each article. This topic model was previously validated for capturing astrobiological themes (Malaterre and Lareau 2023). (2) Author Topic Profiles—For each author, we computed topic profiles by averaging the topic distributions of the articles they contributed to, segmented into four time periods to enable diachronic analysis. (3) Actor Network Construction—Correlation matrices of author topic profiles were used to construct actor networks, where edges represent semantic similarities between authors. (4) Com-

munity Detection—Louvain community detection was applied to identify clusters of authors (Blondel et al. 2008), with each cluster representing an HCoI. Community topic profiles were computed by averaging the topic profiles of their members, allowing straightforward semantic interpretation. And (5) Diachronic Analysis—By measuring distances between community topic profiles across time periods, we mapped the evolution and genealogical relationships of communities.

Results

Our methods revealed several key findings. First, in terms of the identification of hidden communities of interest, the semantic similarities between authors allowed us to infer meaningful clusters of researchers, even in the absence of explicit relational data. These HCoIs reflect underlying thematic commonalities within the astrobiology community (Fig. 1). Second, the community topic profiles provided immediate insights into the thematic focus of each group, thereby facilitating the interpretation of the communities. For example, certain clusters emphasized planetary habitability, while others concentrated on origins-of-life experiments or extremophiles. Finally, diachronic analysis revealed the temporal evolution of communities, including the emergence of new clusters, the fragmentation of existing ones, and thematic shifts over time. This genealogical mapping highlighted the dynamic interplay of scientific interests in astrobiology.

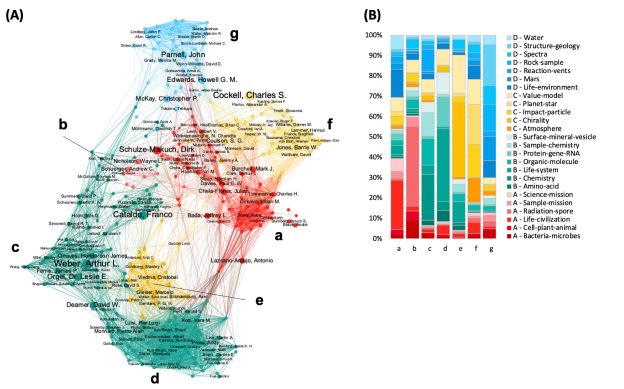

Figure 1. (A) Astrobiology semantic social network 2000-2009 (B) Community topic profiles

Discussion

The approach demonstrates the utility of combining topic modeling and community detection to infer semantic social networks. Compared to traditional SNA methods, this framework allows for the exploration of latent relationships based on shared semantic content, overcoming the limitations posed by the absence of explicit relational data. By focusing on "who talks about the same thing as who", our method captures thematic affinities that often underpin scientific research communities but may not be reflected in citation or co-authorship networks. This provides a complementary perspective on scientific communities, revealing the shared intellectual terrain that connects researchers. Beyond astrobiology, our approach can be applied to various domains where textual data are abundant, but relational data are lacking. Examples include the identification in sociology of networks based on shared narratives in blogs or newsfeeds; literary studies, the analysis of thematic clusters of authors in specific literary genres; in historical studies, the mapping of intellectual communities from archival documents or correspondence; or in philosophy, the investigation of philosophers' thematic communities and intellectual genealogies. By offering a scalable and interpretable framework, this approach opens new avenues for exploring the structure and dynamics of social and intellectual networks.

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